

```
!pip install ultralytics opencv-python matplotlib
```

Collecting ultralytics

Downloading ultralytics-8.4.161-py3-none-any.whl.metadata (46 kB)

46.7/46.7 kB 756.0 kB/s eta 0:00:00

Requirement already satisfied: opencv-python in /usr/local/lib/python3.13/dist-packages (5.0.0.93)

Requirement already satisfied: matplotlib in /usr/local/lib/python3.13/dist-packages (3.10.0)

Requirement already satisfied: cloudpickle>=3.1.1 in /usr/local/lib/python3.13/dist-packages (from ultralytics)

Requirement already satisfied: filelock>=3.16.1 in /usr/local/lib/python3.13/dist-packages (from ultralytics)

Requirement already satisfied: numpy>=1.23.0 in /usr/local/lib/python3.13/dist-packages (from ultralytics)

Requirement already satisfied: pillow>=7.1.2 in /usr/local/lib/python3.13/dist-packages (from ultralytics)

Requirement already satisfied: pyyaml>=5.3.1 in /usr/local/lib/python3.13/dist-packages (from ultralytics)

Requirement already satisfied: requests>=2.23.0 in /usr/local/lib/python3.13/dist-packages (from ultralytics)

Requirement already satisfied: torch>=1.8.0 in /usr/local/lib/python3.13/dist-packages (from ultralytics)

Requirement already satisfied: torchvision>=0.9.0 in /usr/local/lib/python3.13/dist-packages (from ultralytics)

Requirement already satisfied: psutil>=5.8.0 in /usr/local/lib/python3.13/dist-packages (from ultralytics)

Requirement already satisfied: polars>=0.20.0 in /usr/local/lib/python3.13/dist-packages (from ultralytics)

Collecting nvidia-ml-py>=12.0.0 (from ultralytics)

Downloading nvidia_ml_py-13.610.43-py3-none-any.whl.metadata (9.7 kB)

Collecting ultralytics-thop>=2.1.6 (from ultralytics)

Downloading ultralytics_thop-2.1.6-py3-none-any.whl.metadata (13 kB)

Collecting ultralytics-platform>=0.1.45 (from ultralytics)

Downloading ultralytics_platform-0.1.62-py3-none-any.whl.metadata (16 kB)

Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.13/dist-packages (from matplotlib)

Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.13/dist-packages (from matplotlib)

Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.13/dist-packages (from matplotlib)

Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.13/dist-packages (from matplotlib)

Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.13/dist-packages (from matplotlib)

Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.13/dist-packages (from matplotlib)

Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.13/dist-packages (from matplotlib)

Requirement already satisfied: polars-runtime-32==1.35.2 in /usr/local/lib/python3.13/dist-packages (from polars)

Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.13/dist-packages (from python-dateutil)

Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.13/dist-packages (from requests)

Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.13/dist-packages (from requests)

Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.13/dist-packages (from requests)

Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.13/dist-packages (from requests)

Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python3.13/dist-packages (from torch)

Requirement already satisfied: setuptools<82 in /usr/local/lib/python3.13/dist-packages (from torch)

Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.13/dist-packages (from torch)

Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.13/dist-packages (from torch)

Requirement already satisfied: Jinja2 in /usr/local/lib/python3.13/dist-packages (from torch)

Requirement already satisfied: fsspec>=0.8.5 in /usr/local/lib/python3.13/dist-packages (from torch)

Requirement already satisfied: httpx<1,>=0.28 in /usr/local/lib/python3.13/dist-packages (from torch)

Requirement already satisfied: anyio in /usr/local/lib/python3.13/dist-packages (from httpx)

Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.13/dist-packages (from httpx)

Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.13/dist-packages (from httpcore)

Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.13/dist-packages (from sympy)

Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.13/dist-packages (from Jinja2)

Downloading ultralytics-8.4.161-py3-none-any.whl (1.5 MB)

1.5/1.5 MB 28.6 MB/s eta 0:00:00

Downloading nvidia_ml_py-13.610.43-py3-none-any.whl (53 kB)

53.2/53.2 kB 3.3 MB/s eta 0:00:00

Downloading ultralytics_platform-0.1.62-py3-none-any.whl (90 kB)

90.0/90.0 kB 5.6 MB/s eta 0:00:00

Downloading ultralytics_thop-2.1.6-py3-none-any.whl (30 kB)

Installing collected packages: nvidia-ml-py, ultralytics-thop, ultralytics-platform, ultralytics

Successfully installed nvidia-ml-py-13.610.43 ultralytics-8.4.161 ultralytics-platform-0.1.62 ultralytics-thop-2.1.6

```
from ultralytics import YOLO
import cv2
import matplotlib.pyplot as plt
from PIL import Image
```

Creating new Ultralytics Settings v0.0.8 file 

View Ultralytics Settings with 'yolo settings' or at '/root/.config/Ultralytics/settings.json'

Update Settings with 'yolo settings key=value', i.e. 'yolo settings runs_dir=path/to/dir'. For help see

```
model = YOLO("yolov8n.pt")
```

```
print("YOLOv8 Model Loaded Successfully")
```

Downloading <https://github.com/ultralytics/assets/releases/download/v8.4.0/yolov8n.pt> to 'yolov8n.pt':
YOLOv8 Model Loaded Successfully

```
import urllib.request
```

```
url = "https://ultralytics.com/images/bus.jpg"  
urllib.request.urlretrieve(url, "sample.jpg")
```

```
print("Image Downloaded")
```

Image Downloaded

```
img = cv2.imread("sample.jpg")  
img_rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
```

```
plt.figure(figsize=(10,6))  
plt.imshow(img_rgb)  
plt.axis("off")  
plt.title("Original Image")  
plt.show()
```

Original Image



```
results = model("sample.jpg")
```

```
results
```

```

[[123, 150, 176],
 [124, 151, 177],
 [125, 155, 180],
 ...,
 [161, 171, 188],
 [160, 170, 187],
 [160, 170, 187]],

...,

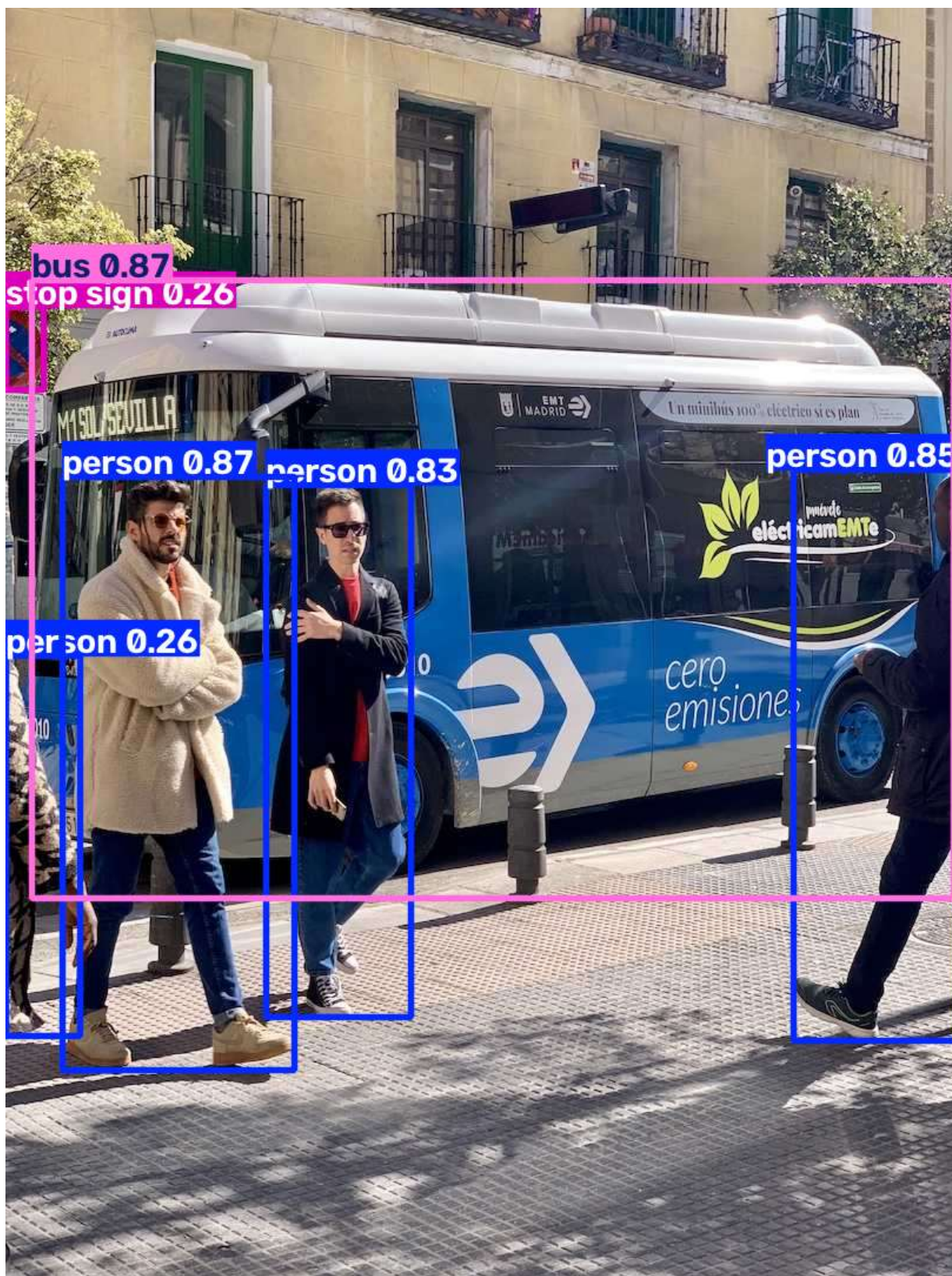
[[183, 182, 186],
 [179, 178, 182],
 [180, 179, 183],
 ...,
 [121, 111, 117],
 [113, 103, 109],
 [115, 105, 111]],

[[165, 164, 168],
 [173, 172, 176],
 [187, 186, 190],
 ...,
 [102, 92, 98],
 [101, 91, 97],
 [103, 93, 99]],

[[123, 122, 126],
 [145, 144, 148],
 [176, 175, 179],
 ...,
 [ 95, 85, 91],
 [ 96, 86, 92],
 [ 98, 88, 94]]], dtype=uint8)
orig_shape: (1080, 810)
path: '/content/sample.jpg'
probs: None
save_dir: '/content/runs/detect/predict'
semantic_mask: None
speed: {'preprocess': 33.786179999999278, 'inference': 523.7123020000354, 'postprocess':
39.64344100006656}]

```

```
results[0].show()
```



```
results[0].save(filename="detected_output.jpg")
```

```
print("Output Saved")
```

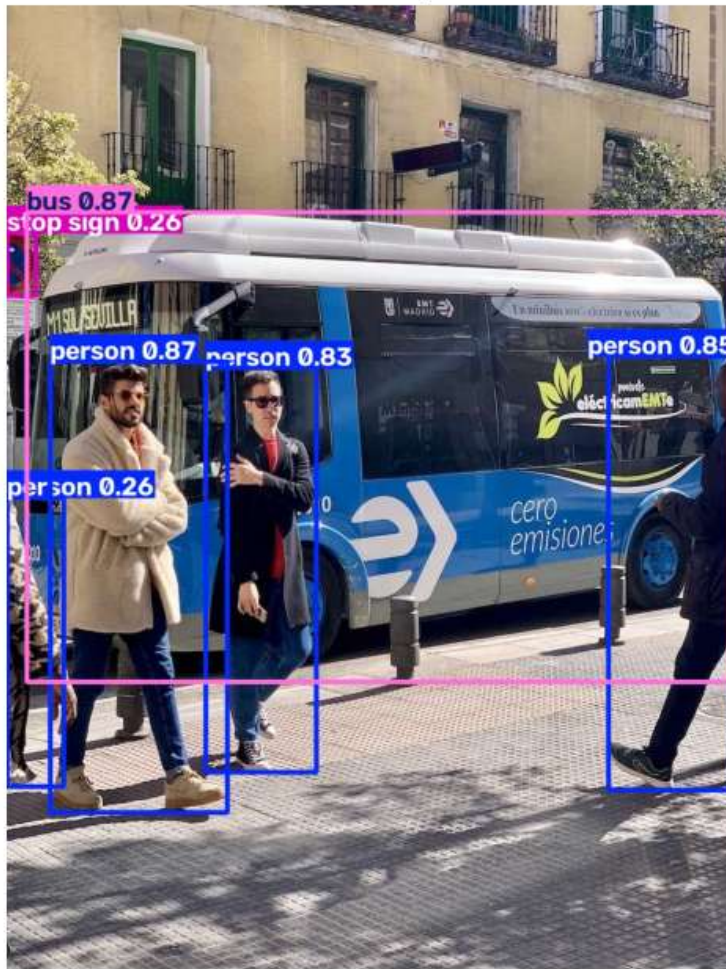
Output Saved

```
output = cv2.imread("detected_output.jpg")
output = cv2.cvtColor(output, cv2.COLOR_BGR2RGB)

plt.figure(figsize=(12,8))
plt.imshow(output)
```

```
plt.axis("off")
plt.title("Detected Objects")
plt.show()
```

Detected Objects



```
for box in results[0].boxes:
    cls = int(box.cls[0])
    conf = float(box.conf[0])

    print(
        f"Object: {model.names[cls]} | Confidence: {conf:.2f}"
    )
```

```
Object: bus | Confidence: 0.87
Object: person | Confidence: 0.87
Object: person | Confidence: 0.85
Object: person | Confidence: 0.83
Object: person | Confidence: 0.26
Object: stop sign | Confidence: 0.26
```

```
from collections import Counter

detected = []

for box in results[0].boxes:
    cls = int(box.cls[0])
    detected.append(model.names[cls])

count = Counter(detected)

print("Detected Objects:")
for obj, num in count.items():
    print(obj, ":", num)
```

Detected Objects:

bus : 1

person : 4

stop sign : 1

```
print("""  
Observation:
```

```
1. YOLOv8 successfully detected multiple objects.  
2. Bounding boxes were generated around detected objects.  
3. Confidence scores indicate prediction reliability.  
4. Detection was performed in real-time using a pretrained model.  
5. The model can be used in surveillance, autonomous driving,  
   healthcare and smart city applications.  
""")
```

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